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Projects

Singularity - AI Deep Research Platform | Mar 2026 – Apr 2026 | singularity.hellonish.dev

FastAPI · ARQ · PostgreSQL 16 · Redis 7 · Docker · AWS · Caddy · GitHub Actions · Next.js 16 · React 19

- Architected 6-container production stack (Caddy, PostgreSQL 16, Redis 7, API server, ARQ background worker, Next.js frontend) on AWS t3-small (2GB RAM, \$26/month); multi-stage Docker builds, per-container healthchecks, orphan job recovery on worker restart; GitHub Actions CI pipeline against real PostgreSQL 16 + Redis 7 service containers; 25,500 LOC shipped solo in 13 days
- Engineered async Python backend: ARQ worker system (4 concurrent, 30-min timeout, 3 retries, idempotency via PostgreSQL partial unique index), `asyncio.gather()` wave parallelism, Redis pub/sub, FastAPI SSE event bus; Redis sliding-window rate limiter (ZREMRANGE + ZCARD + ZADD) in single atomic pipeline transaction, fail-open on Redis outage
- Designed PostgreSQL 16 schema (8 tables, 12 indexes, 5 Alembic migrations) tuned for 2GB VM (`shared_buffers=64MB`, `asyncpg pool_size=5`, `max_overflow=10`, `pool_pre_ping=True`); JWT family rotation with reuse detection (compromised token revokes entire family), Fernet-encrypted secrets at rest, 30-second single-use SSE JWT, non-root Docker container, Caddy auto-SSL

Wand - AI Career Intelligence Platform | Jan 2026 – Apr 2026

Redis · WebSockets · asyncpg · SQLAlchemy · SQLite · PyMuPDF · trafilatura · Docker

- Implemented dual execution architecture: Celery + Redis broker for traditional analysis pipeline (qualification → formatting → keyword match) and FastAPI BackgroundTasks + asyncio for 6-step LLM pipeline - both publishing to same Redis pub/sub channel, appearing as identical WebSocket event types on frontend; blocking LLM calls offloaded to executor via `loop.run_in_executor()`
- Built multi-source document ingestion (PDF via PyMuPDF, HTML via trafilatura, structured JSON via instructor) with semantic entity deduplication and merge-priority rules; SQLite JSON columns for evolving AI-generated schemas; additive column migrations via `ensure_sqlite_schema()` at startup; short-lived `SessionLocal()` sessions per write to avoid holding connections across long async tasks
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Snap2Caption — ML Systems for Caption Generation | Mar 2025 – May 2025 | [github](https://github.com)

Terraform · Ansible · Kubernetes (Kubespray) · Docker · MLflow · Prometheus · Grafana · FastAPI · MinIO · rclone

- Built full IaC stack: Terraform (3-VM provisioning on Chameleon Cloud), Ansible (SSH config, Docker install, playbook-driven setup), Kubernetes deployed via Kubespray; block storage (KVM@TACC) for MLflow/MinIO/Prometheus/Grafana/PostgreSQL state; object storage (CHI@UC) for 100k-image training dataset mounted via rclone FUSE
- Deployed FastAPI inference server (<2s P90 latency, ~300 req/hr peak) with Prometheus metrics endpoint and Grafana dashboard; feedback loop: user corrections, MinIO, Label Studio annotation, MLflow retraining artifact registration; ETL pipeline (aria2 extract, city-filter transform, Chameleon object store load); 80/10/10 data split

Finassistant — Multi-Agent Financial Research Platform | Nov 2025 – Dec 2025 | [github](https://github.com)

FastAPI · WebSockets · MongoDB · ChromaDB · LangGraph · React 18 · Docker · AWS EC2 · S3 · CloudFront

- Built event-driven async FastAPI backend with WebSocket token-level streaming; MongoDB-backed session management (multi-user, session persistence, message metadata); dual Docker Compose configurations for full-stack local, backend-only EC2, and frontend-only testing; AWS EC2 deployment with Nginx reverse proxy + Let's Encrypt SSL; React 18 (Vite) on S3+CloudFront CDN

Work Experience

Machine Learning Teaching Assistant | NYU Tandon School of Engineering | NY, USA | Sep 2025 - Present

- Standardized PyTorch training pipelines (data loading, training loops, evaluation) used by 100+ students across course offerings
- Led debugging sessions on CNN optimization, backpropagation, autograd; created supplementary video lectures on ML calculus concepts for graduate cohort

Co-Founder | Macverin Technologies | www.macverin.com | Jul 2022 - Aug 2024

- Architected distributed microservices (FastAPI + Node.js) serving 1K+ DAU for 10+ clients; maintained 99.9% system availability with automated CI/CD pipelines and AWS deployment
- Built full-stack products end-to-end: TypeScript/React/Next.js frontends, ML inference backends; managed complete SDLC from UI/UX design (Figma) to production cloud deployment

Education

- **New York University - Tandon School of Engineering** | New York, US | Sep 2024 - Expected May 2026
Master of Science in Computer Engineering · GPA: 3.67
FOCUS: LANGUAGE MODELS, MULTIMODAL MODELS, TRAINING & FINETUNING
- **Dr. A.P.J. Abdul Kalam Technical University** | Delhi, India | Aug 2020 - Jun 2024
B.Tech Computer Science and Engineering · 1st Division with Distinction

Technical Skills

- **Backend:** FastAPI, PostgreSQL, Redis, ARQ, asyncpg, SQLAlchemy (async), Celery, WebSockets, SSE, REST, gRPC, Node.js, Django, Express
- **Infra:** Docker, Kubernetes, AWS (EC2, S3, RDS, IAM, CloudFront), Terraform, Ansible, GitHub Actions, Prometheus, Grafana, Caddy, Nginx
- **AI/ML:** RAG pipelines, LangGraph, LLM fine-tuning (LoRA/QLoRA), Vector stores (Qdrant, Chroma), fastembed/ONNX, MLflow, PyTorch, Instructor (structured outputs)
- **Frontend:** Next.js, React, TypeScript
- **Languages:** Python, TypeScript, Java, C++, SQL